

Bridge Test and Redesign Evidence

NAME _____ DATE _____ PERIOD _____

Fixed Samples — Predict Before the Reveal

The three fictional classroom samples use the same staged load and stop rule. Study only the visible systems before you predict.

SAMPLE	VISIBLE SYSTEM
A	Flat beam; two main rails; taped card deck
B	Triangulated side frames; cross-braced deck
C	Curved/segmented sides; narrow support contact

Before revealing the results, predict which sample will complete the highest stage and cite one visible clue:

Optional demonstration boundary: A teacher may demonstrate one prepared model only after a tested kit, secured supports, catch tray, marked keep-clear zone, known lightweight stages, safe cap, objective stop rule, and workable class schedule are ready. Students do not place loads. Never use textbooks or unstable desks.

Reveal and Compare

Your teacher reveals this fictional result card after everyone records a prediction.

SAMPLE	MAXIMUM COMPLETED STAGE	STOP EVIDENCE
A	2	Center deck reached displacement mark before Stage 3
B	5 safe cap	No detached joint or displacement stop at cap
C	3	One support contact slipped before Stage 4

Sample I analyzed: [A / B / C]

Maximum completed stage: [_____] Stop evidence: [_____]

First failed or limited element: [_____]

Which comparison is fair, and what variable is still uncontrolled?

One pattern and one exception across the three results:

Word bank: stage · stop evidence · displacement · support contact · variable · limitation

Individual Failure Analysis and Redesign

Result evidence I used:

One specific redesign:

Why that change should affect the result:

What I would measure in the next test:

Use this frame: Sample [code] reached [stage/result]. I would change [specific feature] because [evidence]. In the next test, I would measure [measure].

Career Role and Real-World Limits

Worker who would review this result next:

Additional evidence a real project would require:

One limit of the classroom prototype:

Use this frame: A [role] would review [evidence] next, but a real project would also require [additional evidence].
This classroom model cannot prove [limit].